Fiscal Year:	FY 2005	**Task Last Updated:**	FY 03/10/2005
PI Name:	Hulet, Randall G. Ph.D.		
Project Title:	Superfluid Phase Transition in an Ultracold Fermi Gas		
Division Name:	Physical Sciences		
Program/Discipline:	FUNDAMENTAL PHYSICS		
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Laser cooled atomic physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77251	**Congressional District:**	9
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	00-HEDS-02
Start Date:	05/01/2001	**End Date:**	04/30/2006
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	2
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	4	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NAG3-2894		
Performance Goal No.:	1.24		
Performance Goal Text:	To explore the frontiers of physics and unlock the mysteries of the universe obscured by Earth's gravity.		
Task Description:	Atoms have been observed to undergo Bose-Einstein condensation when the temperature is reduced to ultralow values. These atoms have integer spin and are bosons. An entirely different class of atoms, the fermions, do not exhibit this phase transition. However, fermions are predicted to form Cooper pairs, in analogy with the electronic pairing responsible for superconductivity, at very low temperatures, and to form a superfluid. We are applying the technology used to produce Bose-Einstein condensation to achieve Cooper pairing in a gas of fermions, namely lithium-6 atoms. The primary objective of this research is to detect a BCS-like phase transition in an ultracold gas of lithium-6 atoms. This type of phase transition requires that particles pair. Pairin is responsible for the phenomena of superconductivity in solids and superfluidity in liquid helium-3. It has never been observed in the gas phase, however, because densities are usually too low for pairs to form and the temperatures to high for the manifestation of quantum statistical effects. In a previously funded NASA project, we showed that lithium-6 atoms exhibit an extraordinarily large attraction for each		

Task Description:	other, which is, in fact, larger than that of any known atom. Even for lithium-6, however, the critical temperature will be ~100 nano-Kelvin. In the past few years, the techniques of laser cooling combined with evaporative cooling in a magnetic trap have been demonstrated to be effective at reaching these low temperatures in gases of bosonic atoms, where they undergo Bose-Einstein condensation. Ultracold trapped atomic gases are remarkably versatile systems for investigations of many-body quantum effects, such as Bose-Einstein condensation. Unlike condensed matter systems, many of the key parameters, such as density, interaction strength, number, and temperature are under experimental control. This ability may enable us to explore the phenomena of Cooper pairing and fermion superfluidity under conditions never before achieved. Of particular significance, is the possibility of achieving Cooper pairing in the strong coupling regime, which has not been observed experimentally, nor has it been understood theoretically.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The results of this research will lead to a greater understanding of the collective behaviour of Fermi systems. This is especially important in the case of superconductors. Despite 20 years of intense effort, the exact mechanism of high-temperature superconductivity has yet to be discovered. We hope that the work that we are doing will provide insight into these important sytems.
Task Progress:	A major goal was accomplished this year. We succeeded in producing a Bose-Einstein condensate (BEC) of molecules made from pairs of lithium-6 atoms. This is particularly significant because lithium-6 is a fermion particle, which themselves are unable to multiply occupy a single quantum state, which is the hallmark of a BEC. The BEC of these molecules can be considered to be a condensation of fermion pairs. Fermion pairs, also known as Cooper pairs, are responsible for superconductivity and superfluidity of helium-3 atoms. In our case, the pairs are electrically neutral so they form a superfluid rather than a superconductor. The interactions between atoms and molecules in the gas are tunable by use a Feshbach resonance, in which pairs of atoms are magnetically tuned to resonance with a molecular bound state. The Feshbach resonance enables us to probe the fundamental nature of Fermi superfluidity as a function of interaction strength. Most importantly, it provides a system in which the interactions are tunable from the weakly interacting (BCS) regime where the atom pairs are spatially large in comparison to the interparticle separation, to the strongly interacting regime where the pairs resemble molecules. We have developed a technique for measuring the molecular content of the pairs by using an optical spectroscopic probe. The probe laser frequency is tuned between the many-body pair states and an electronically excited molecular state. The probe only couples to the molecular part of the pairs, and excitation of a molecule results in a detectable loss of atoms. By performing this experiment as a function of magnetic field, the molecular content may be measured across the BEC/BCS crossover. We have recently discovered that, contrary to expectations, there is a substantial molecular component on the so-called BCS, or “non-molecular” side of the resonance.
Bibliography Type:	Description: (Last Updated: 02/07/2007)
Articles in Peer-reviewed Journals	Strecker, K.E.; Partridge, G.B.; Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Long-Lived Molecular Bose Gas" Physical Review Letters 91, 080406 , Aug-2003
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Articles in Peer-reviewed Journals	Strecker, K.E.; Partridge, G.B.; Truscott, A.G.; Hulet, R.G. "Tunable Interactions in Ultracold Bose Gases" Advances in Space Research (in press), doi:10.1016/j.asr.2003.09.058 , Jul-2004
Awards	Hulet, R.G. "NASA Exceptional Scientific Achievement Medal" Apr-2004
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Books/Book Chapters	K. E. Strecker, G. B. Partridge, R. I. Kamar, and R. G. Hulet "“Conversion of an Atomic Fermi Gas to a Molecular Bose-Einstein Condensate”" in "Atomic Physics 17, Proceedings of the Nineteenth International Conference on Atomic Physics" Ed. V. Bagnato and L. Marcassa American Institute of Physics Conference Proceedings, May-2005
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Dissertations and Theses	Partridge, G.B. "An Improved System for Creating Ultracold Fermi Gases of 6Li" Masters Thesis, Rice University Sep-2003
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Presentation	Hulet, R.G. "“Cold Fermions, Feshbach Resonances, and Molecular Condensates“" Boulder Condensed Matter Summer School, Boulder Jul-2004
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Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Molecular Bose Einstein Condensate" Physics Colloquium, Univ. of California, Berkeley Sep-2004

Presentation	Hulet, R.G. "Molecular Probe of the BCS/BEC Crossover" Center for Ultracold Atoms Seminar, Harvard University Oct-2004
Presentation	Hulet, R.G. "Tunable Interactions in Bose and Fermi Gases: From Molecular BEC's to Solitons" Workshop on Solitons in BEC's, Almagro, Spain Feb-2005
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Presentation	Hulet, R.G. "Strongly Interacting Fermi Gas" Quantum Fluids and Solids Annual Meeting, Albuquerque Aug-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Long-Lived Molecular Bose Gas" European Workshop on Bose-Einstein Condensation, San Feliu, Spain Sep-2003
Presentation	Hulet, R.G. "Conversion of an Atomic Fermi Gas to a Gas of Molecular Bosons" Annual Meeting on the Physics of Quantum Electronics, Snowbird, UT Jan-2004
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